

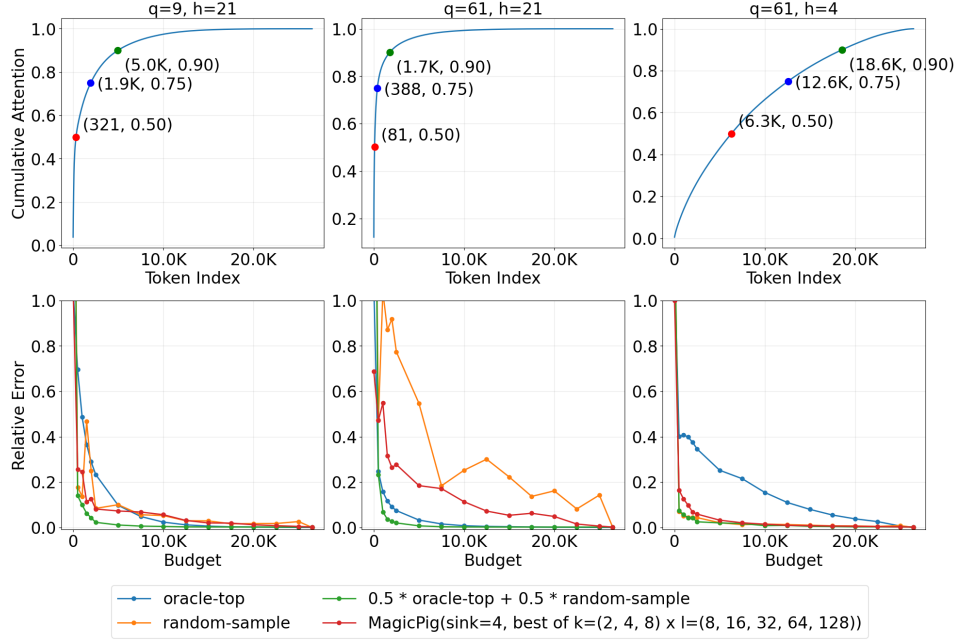


Figure 2: Top pane: cumulative sum of attention scores sorted in descending order of magnitude, showing the number of tokens required to achieve a $p \in (0, 1)$ coverage over the scores. Bottom: relative local attention errors across token budgets, indexed by head h and query q index

3 BACKGROUND AND MOTIVATION

We begin by describing the three categories of attention computation considered in this paper. For clarity, we restrict our exposition to the case of batch size one with a single query vector. Consider KVCache $K, V : n \times d$ and query vector $q : 1 \times d$.

Full Scaled Dot Product Attention (SDPA)

$$\text{SDPA}(\mathbf{K}, \mathbf{V}, \mathbf{q}) = \sum_{i=1}^n (a_i \mathbf{V}[i]) \text{ where } a_i = \frac{\exp \langle \mathbf{K}[i], \mathbf{q} \rangle}{\sum_{j=1}^n \exp \langle \mathbf{K}[j], \mathbf{q} \rangle} \quad (1)$$

where a_i are referred to as attention scores. This represents full attention computation.

Sparse Attention with deterministic index selection: Let S denote the sequence of indices selected by a deterministic method—such as attention sinks, sliding window, top- k selection, or a combination of them. The sparse attention computation based on this deterministic index set is given by:

$$\text{SDPA}_S(\mathbf{K}, \mathbf{V}, \mathbf{q}) = \sum_{i \in S} (\hat{a}_i \mathbf{V}[i]) \text{ where } \hat{a}_i = \frac{\exp \langle \mathbf{K}[i], \mathbf{q} \rangle}{\sum_{j \in S} \exp \langle \mathbf{K}[j], \mathbf{q} \rangle} \quad (2)$$

Sparse Attention with randomized index selection: Let S denote the sequence of indices selected by a randomized method—such as random sampling or MagicPig, and let P be the corresponding sequence of selection probabilities. Given S and P , the attention computation is defined as:

$$\text{SDPA}_{S,P}(\mathbf{K}, \mathbf{V}, \mathbf{q}) = \sum_{(i, p_i) \in (S, P)} (\hat{a}_i \mathbf{V}[i]) \text{ where } \hat{a}_i = \frac{\frac{1}{p_i} \exp \langle \mathbf{K}[i], \mathbf{q} \rangle}{\sum_{(j, p_j) \in (S, P)} \frac{1}{p_j} \exp \langle \mathbf{K}[j], \mathbf{q} \rangle} \quad (3)$$

This definition subsumes deterministic index selection (where probabilities associated are 1) and can be used to represent a selection that is a combination of deterministic and randomized selection.

We motivate our approach through a simple ablation. We use the following baseline: **oracle-top**: selects the tokens with the highest inner products, constrained by the budget, and uses deterministic attention computation. **random-sample**: uniformly samples a subset of tokens (without replacement)